Post-Processing and Glacier Change Analysis

Swun Wunna Htet

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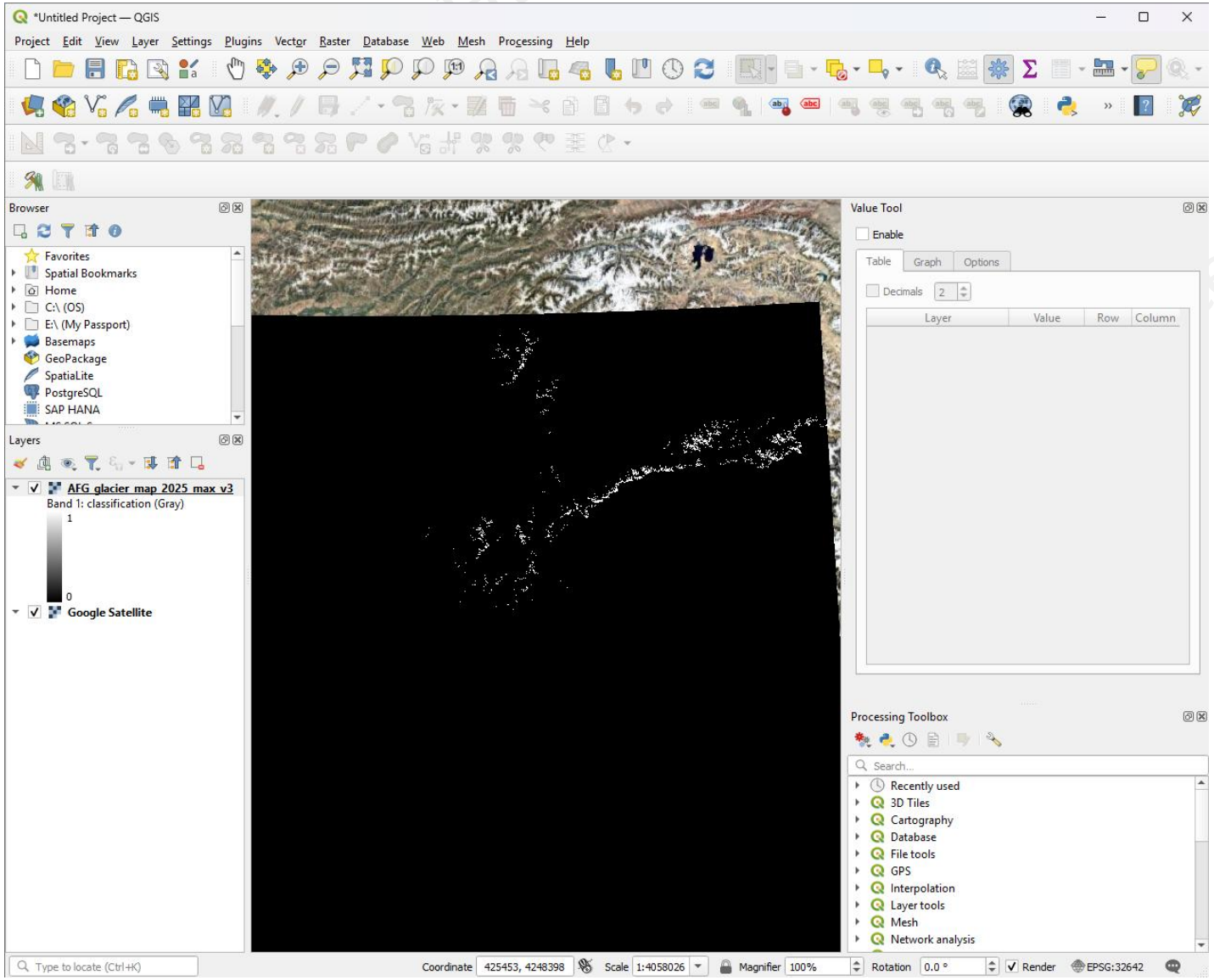
Overview

- This material covers how to covert glacier maps obtained from Google Earth Engine into vectorized glacier inventory using QGIS.

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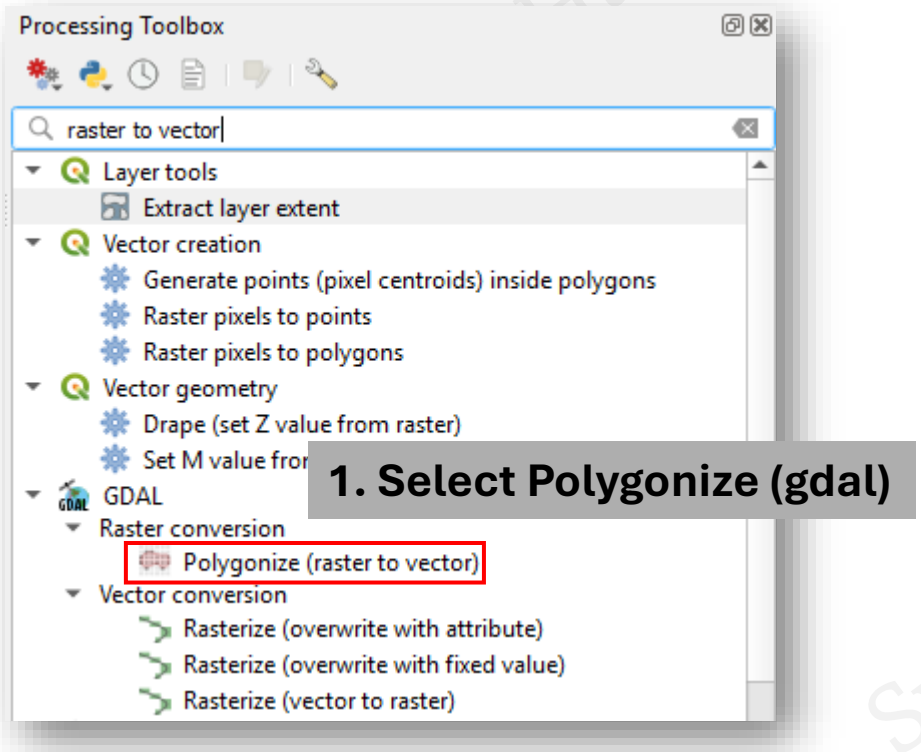
Open QGIS and upload glacier map from GEE

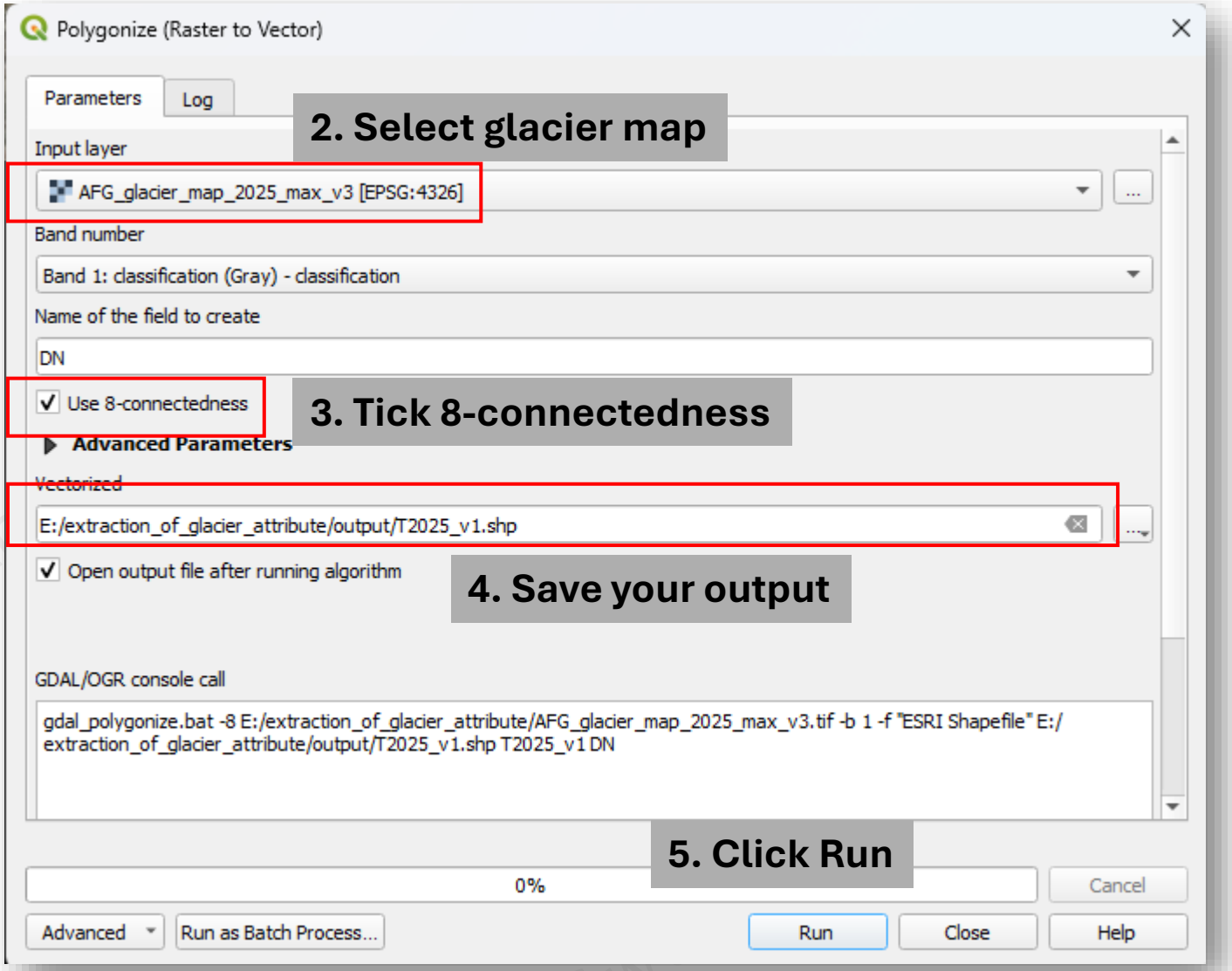


- Filename = select your glacier map.
- We need to vectorize this map.

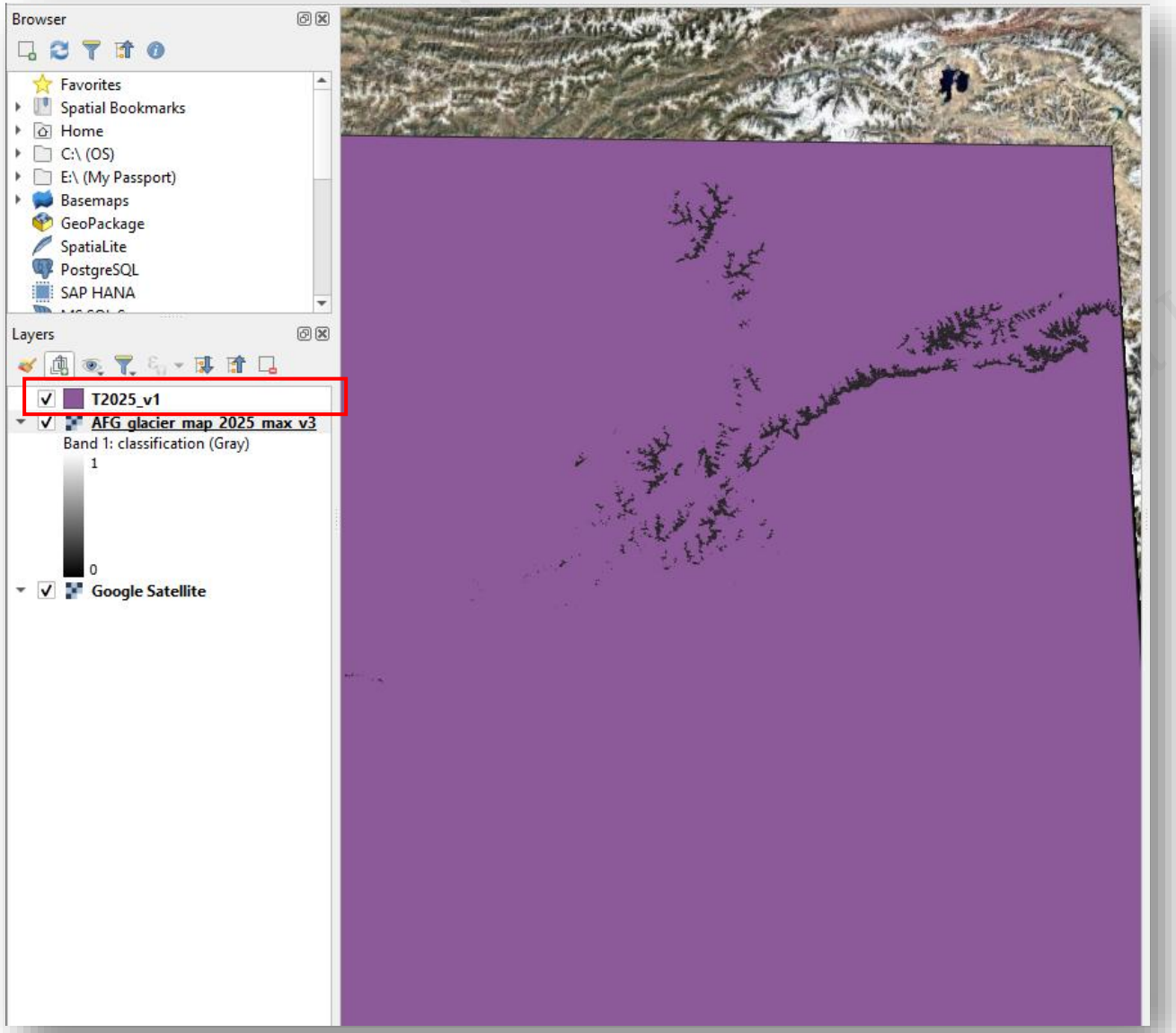
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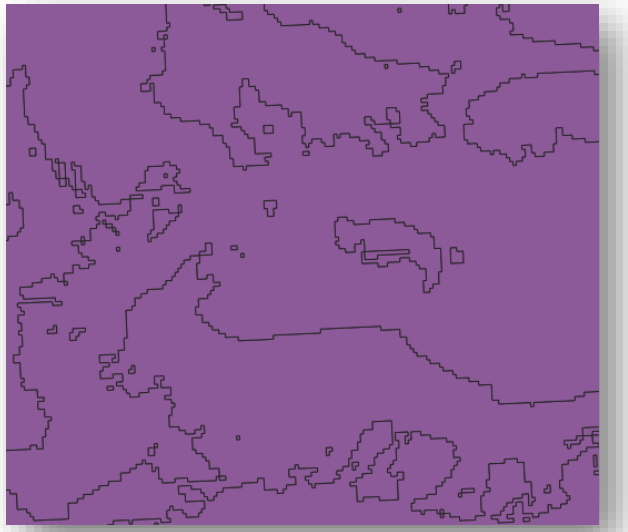
Raster to Vector Operation





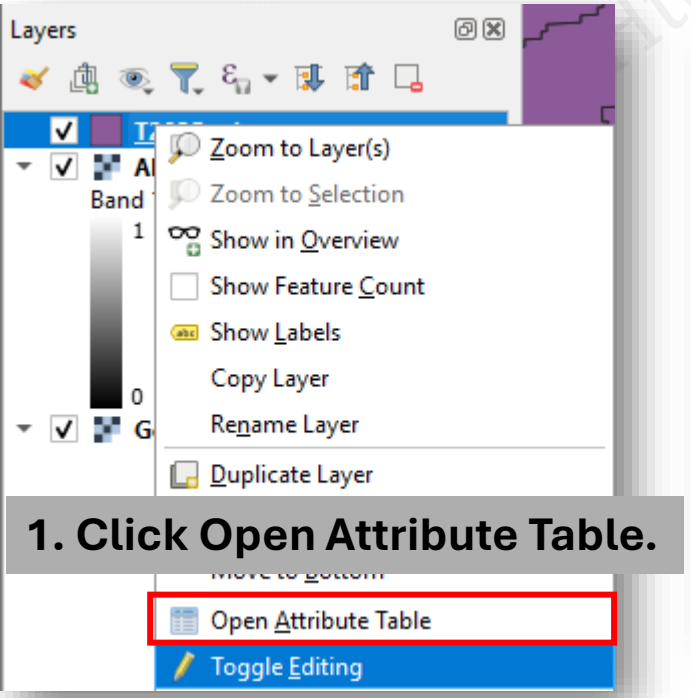
We got the vector boundary



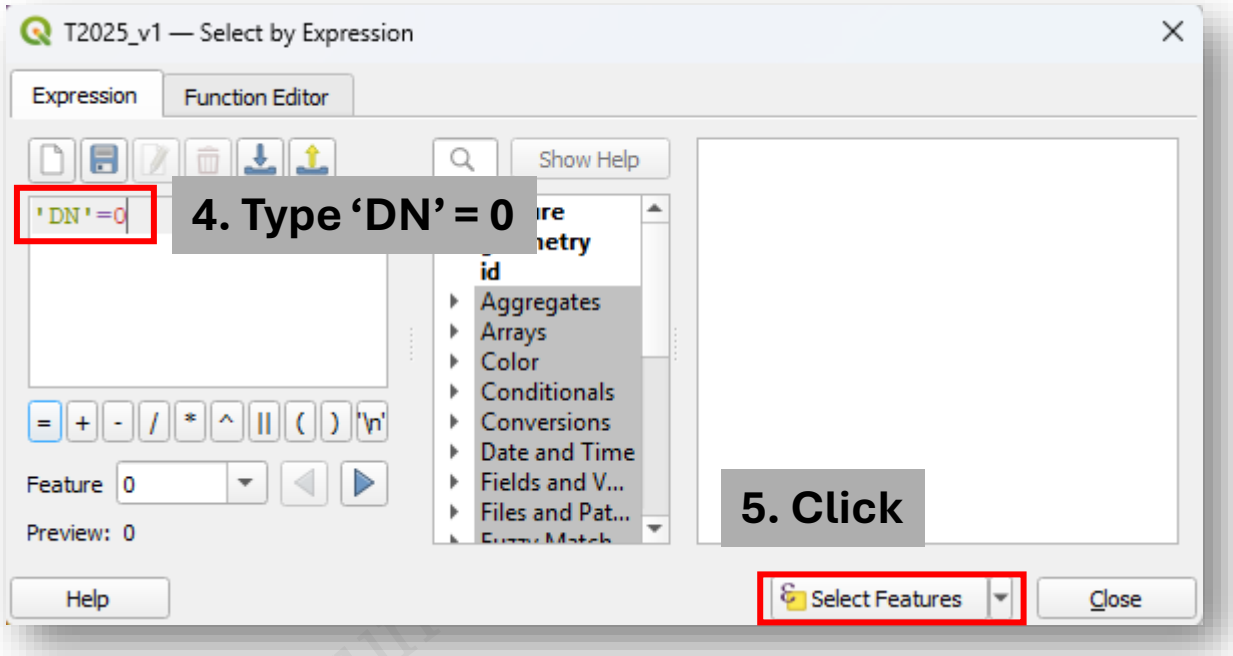


- Notice that jagged boundary due to direct conversion from raster.
- Vector smoothening is required. But before that we need to eliminate non—glacier polygons using attribute selection.

Removing non-glacier polygons

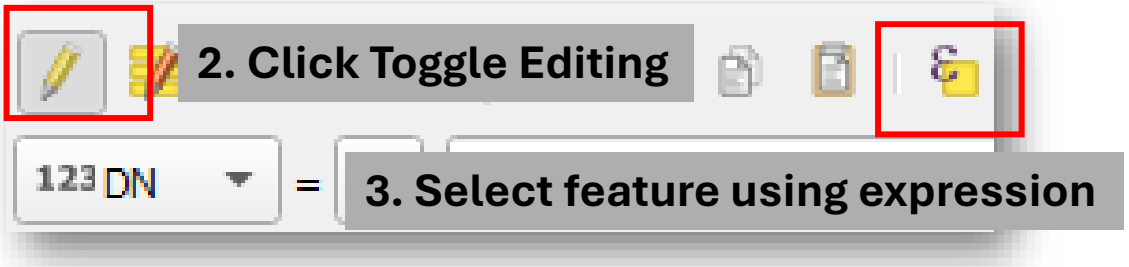


1. Click Open Attribute Table.



4. Type 'DN' = 0

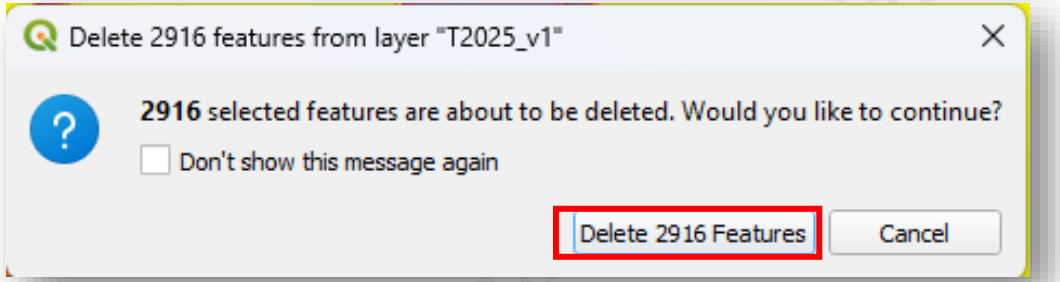
5. Click



2. Click Toggle Editing

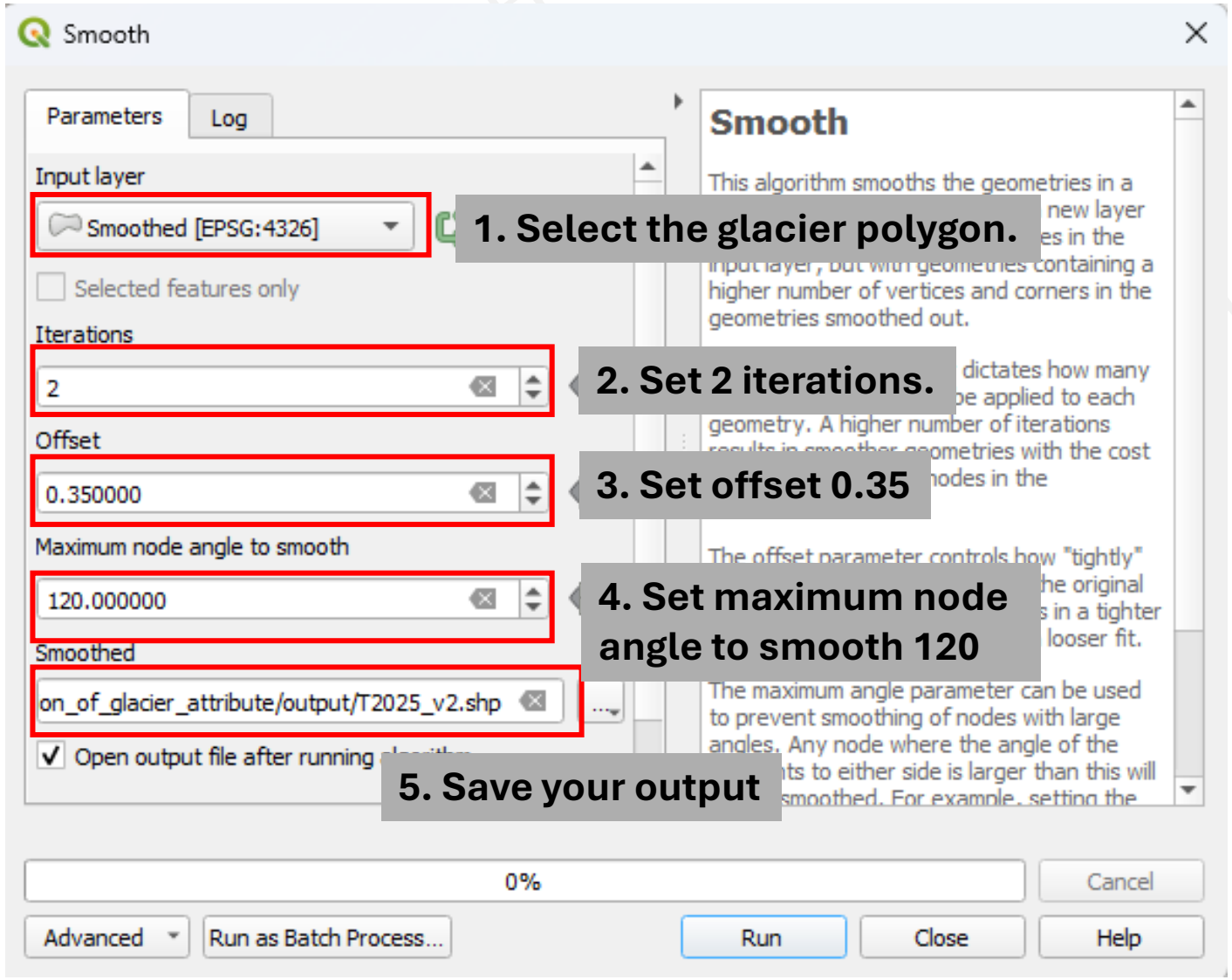
3. Select feature using expression

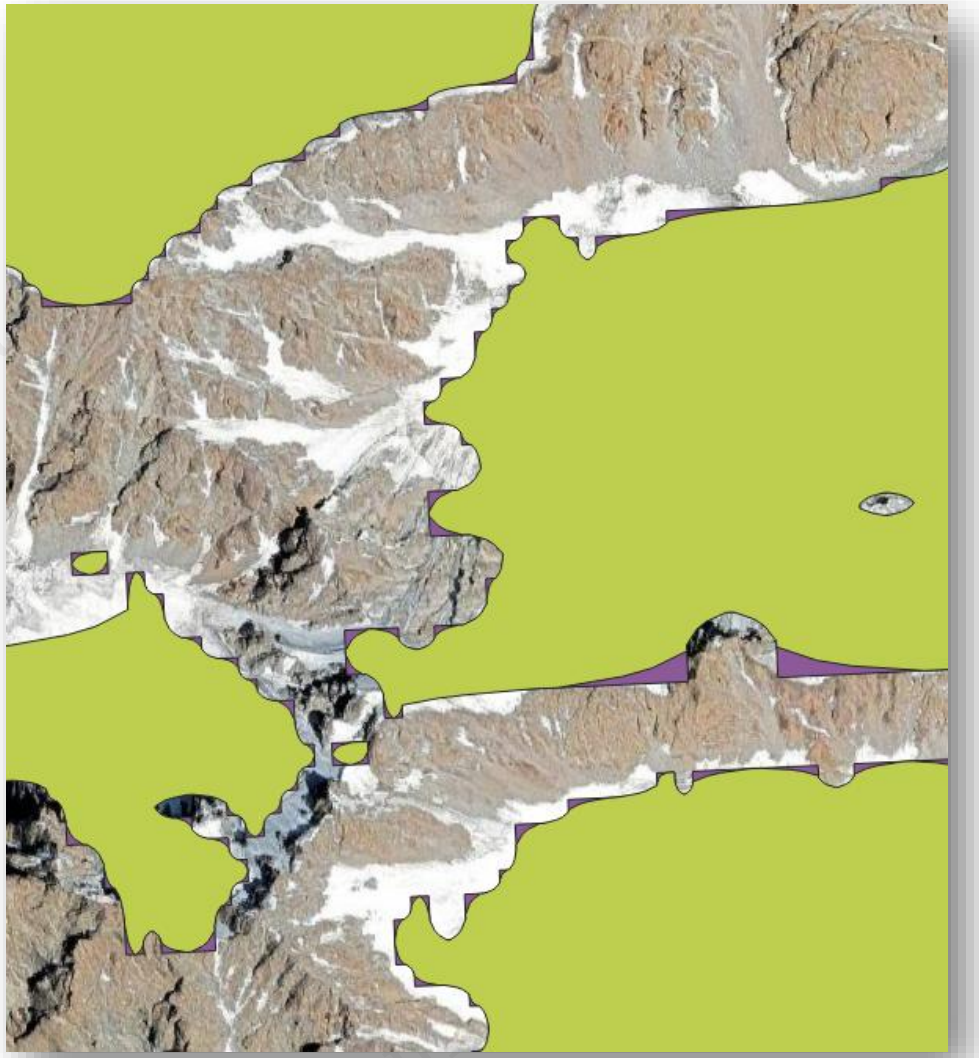
6. After selection, click on Delete selected features.



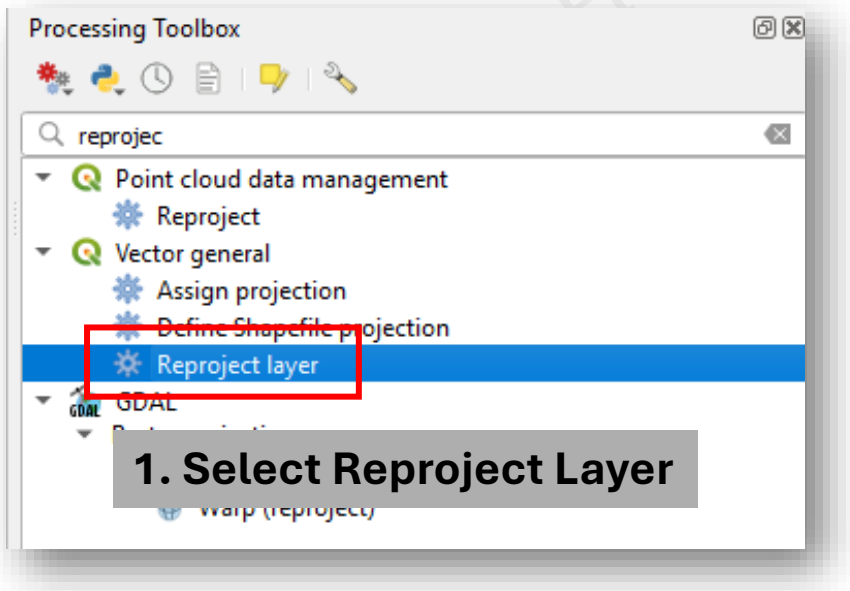
7. Save the editing and close toggle editing mode.

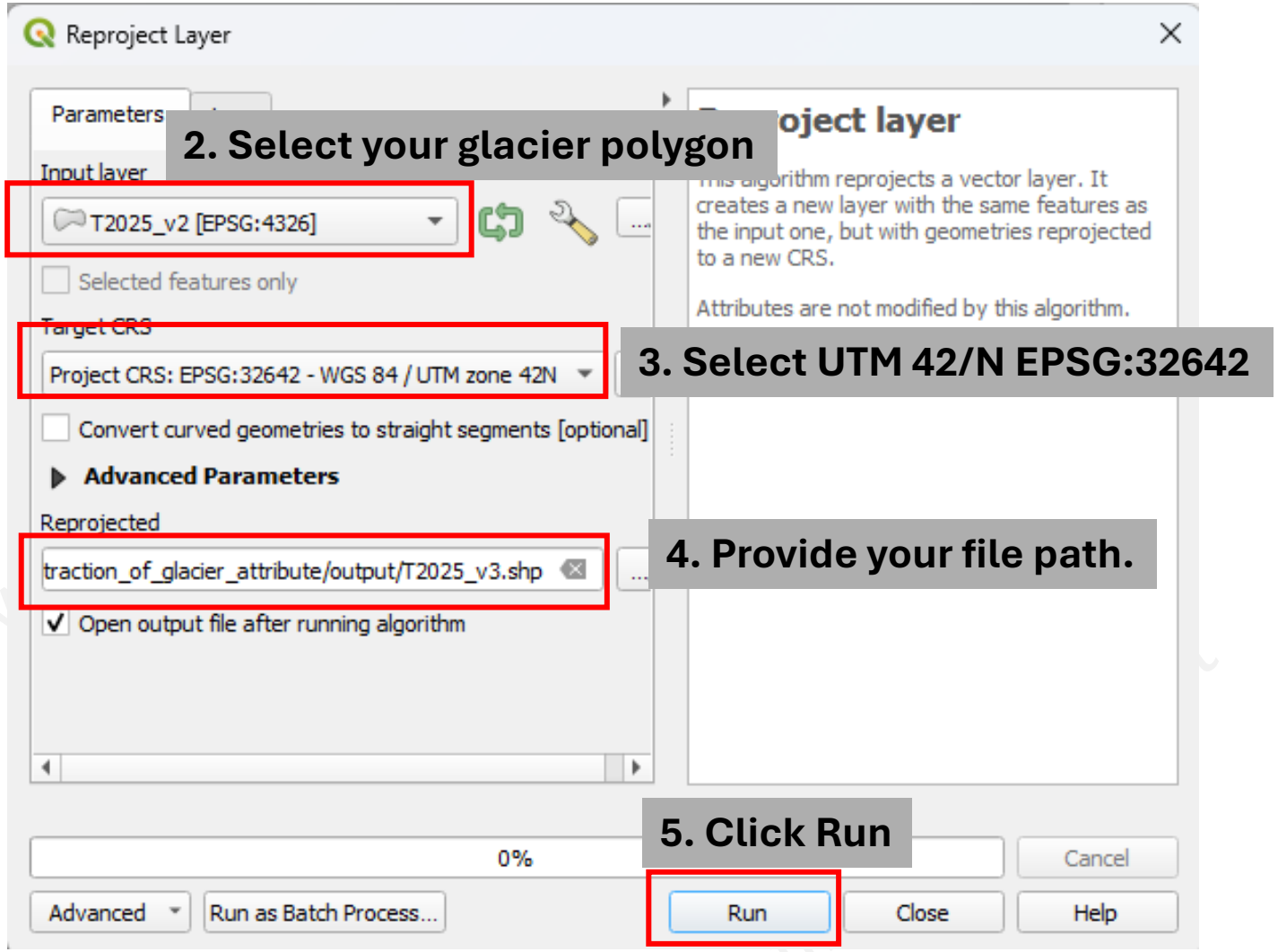
Smoothening the polygon





Reprojection of glacier boundary to UTM 42N Zone



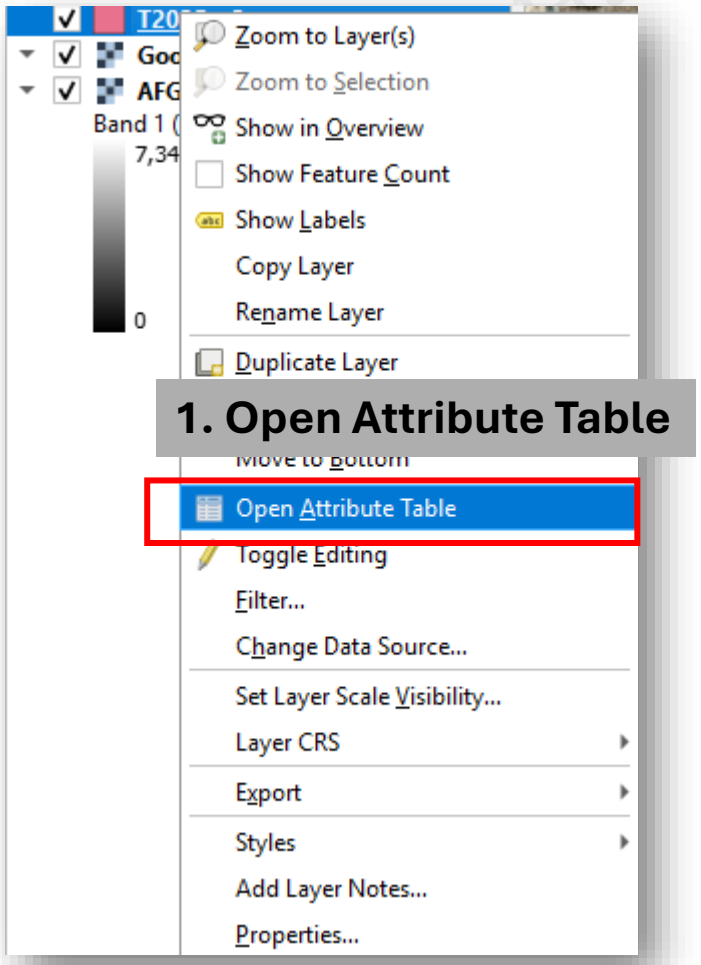


Inventory Attributes

- This section will cover only selected attributes of a proper glacier inventory (NOT ALL which can vary country-to-country).

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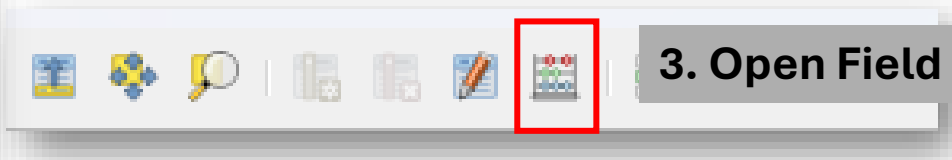
Unique Glacier ID



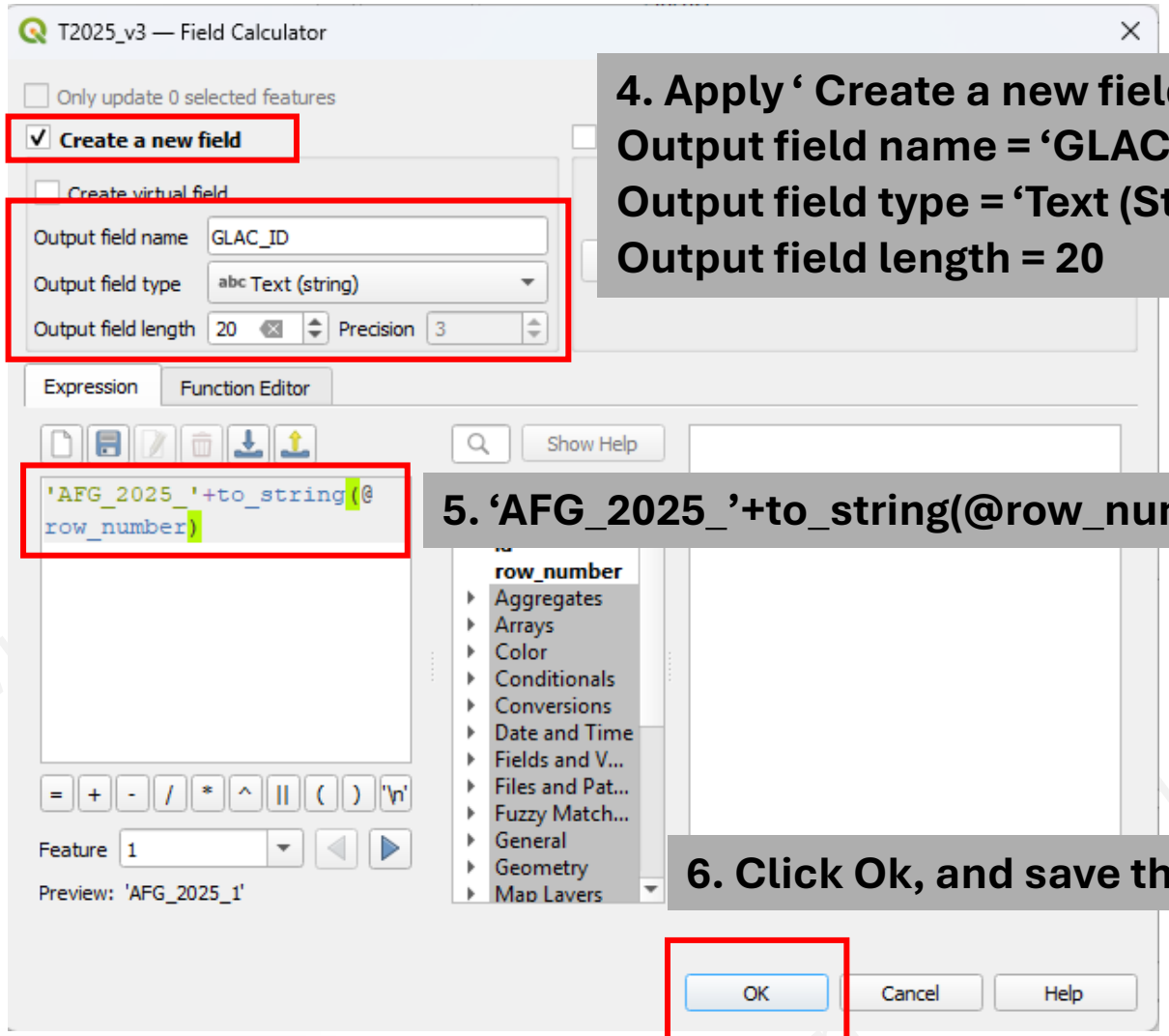
1. Open Attribute Table



2. Activate Toggle Edit mode.



3. Open Field Calculator

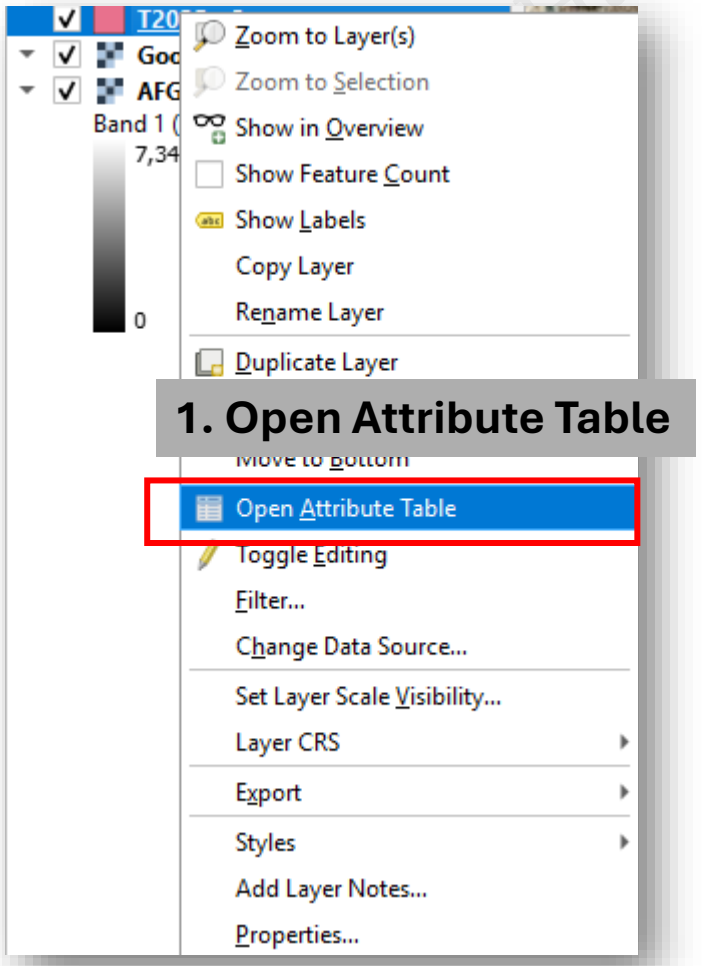


4. Apply ‘ Create a new field’
Output field name = ‘GLAC_ID’
Output field type = ‘Text (String)’
Output field length = 20

5. ‘AFG_2025_’+to_string(@row_number)

6. Click Ok, and save the edit.

Geodetic Area



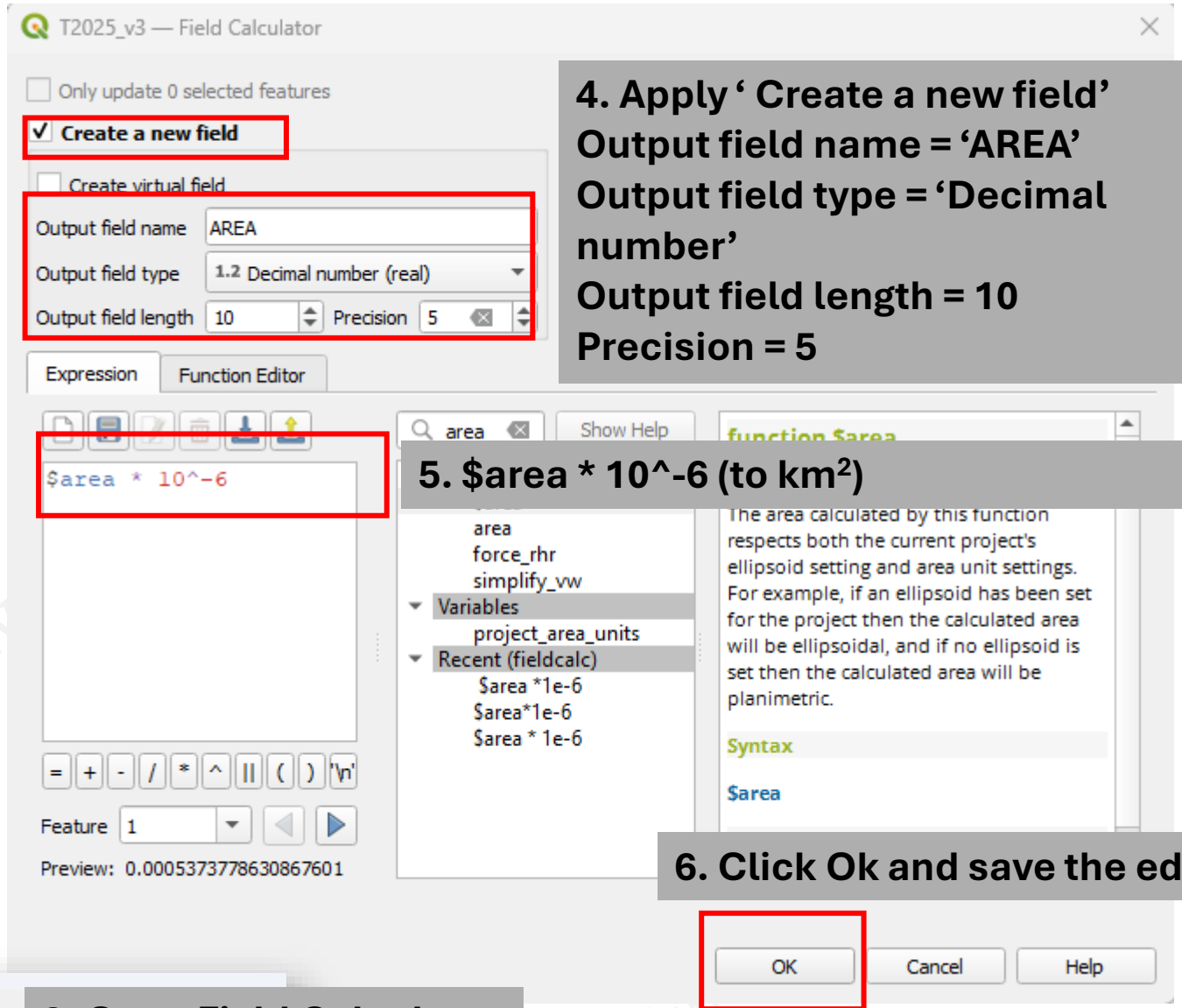
1. Open Attribute Table



2. Activate Toggle Edit mode.



3. Open Field Calculator

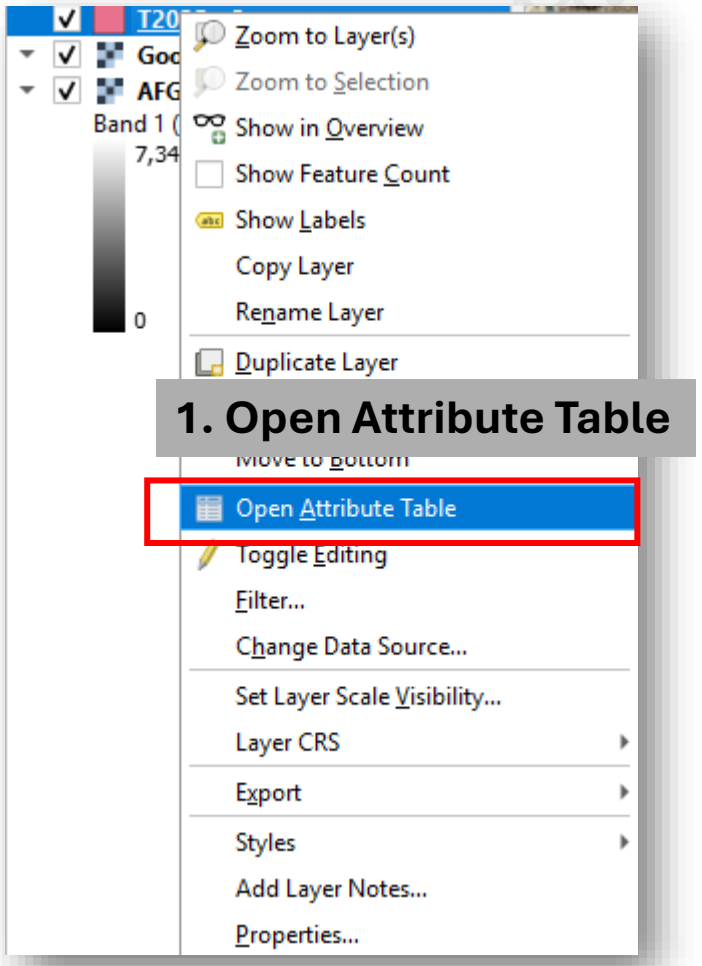


4. Apply ‘ Create a new field’
Output field name = ‘AREA’
Output field type = ‘Decimal number’
Output field length = 10
Precision = 5

5. $area * 10^-6 (to km^2)

6. Click Ok and save the edit.

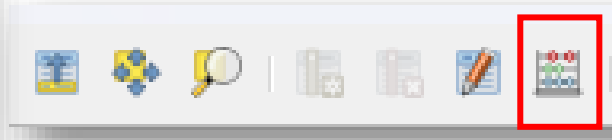
Year



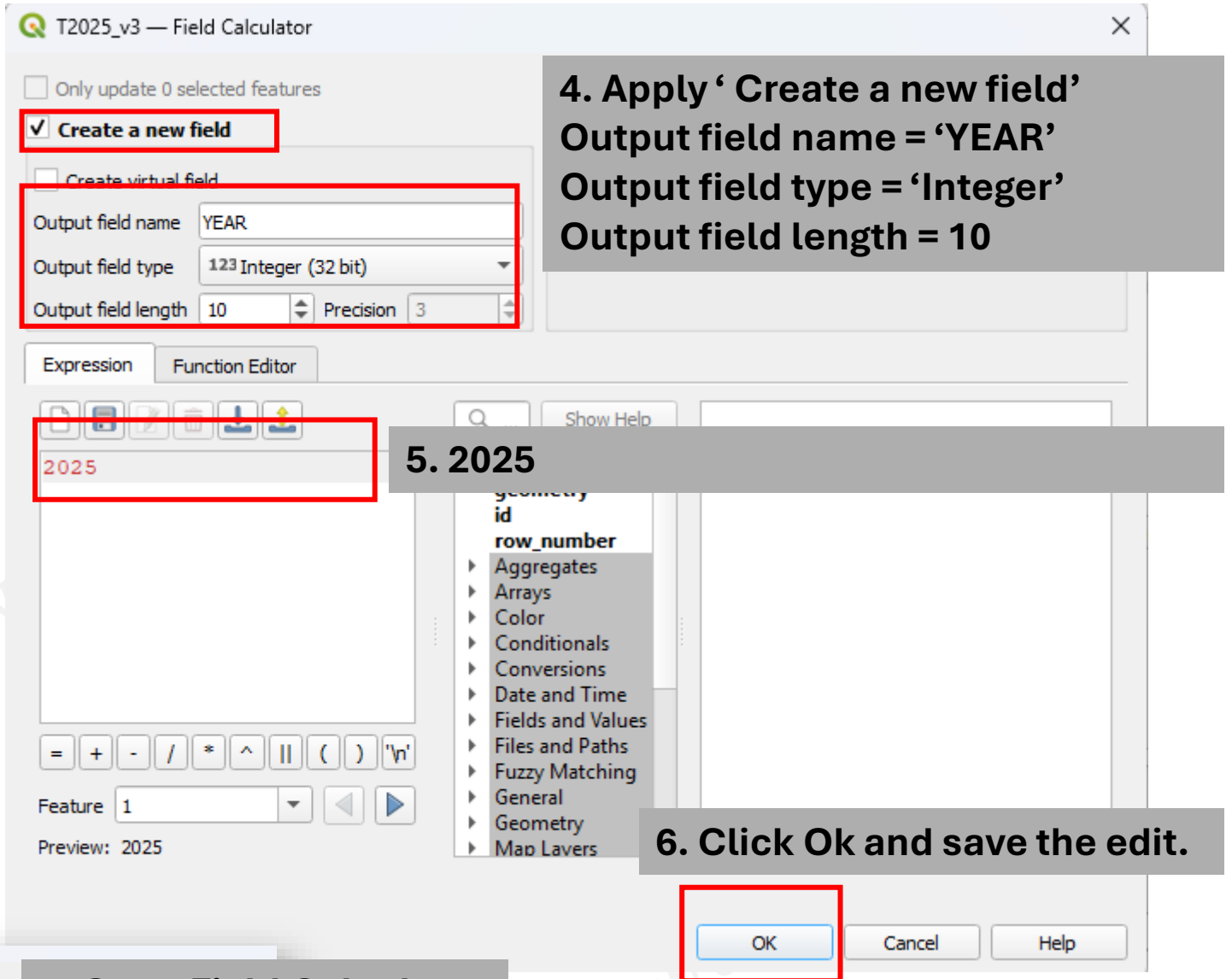
1. Open Attribute Table



2. Activate Toggle Edit mode.



3. Open Field Calculator

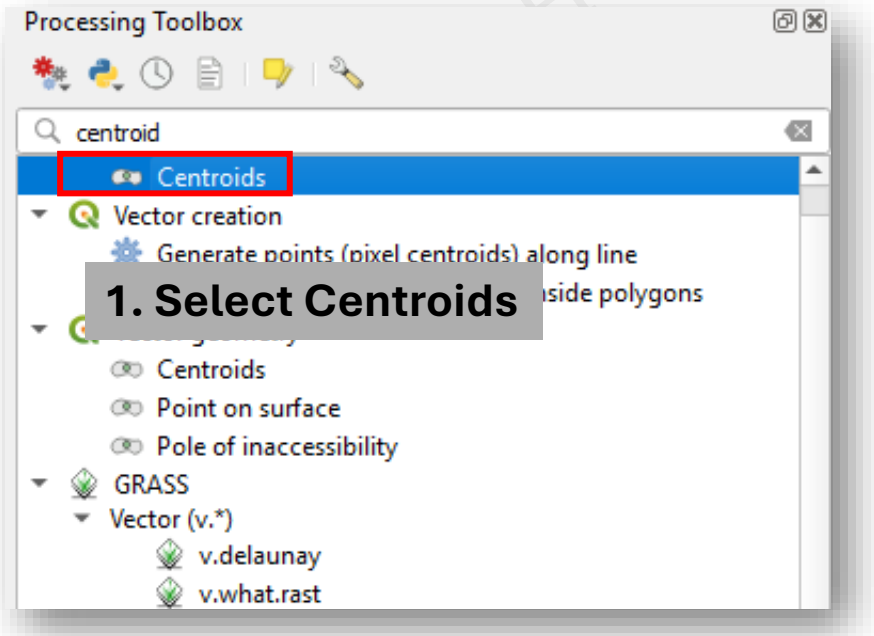


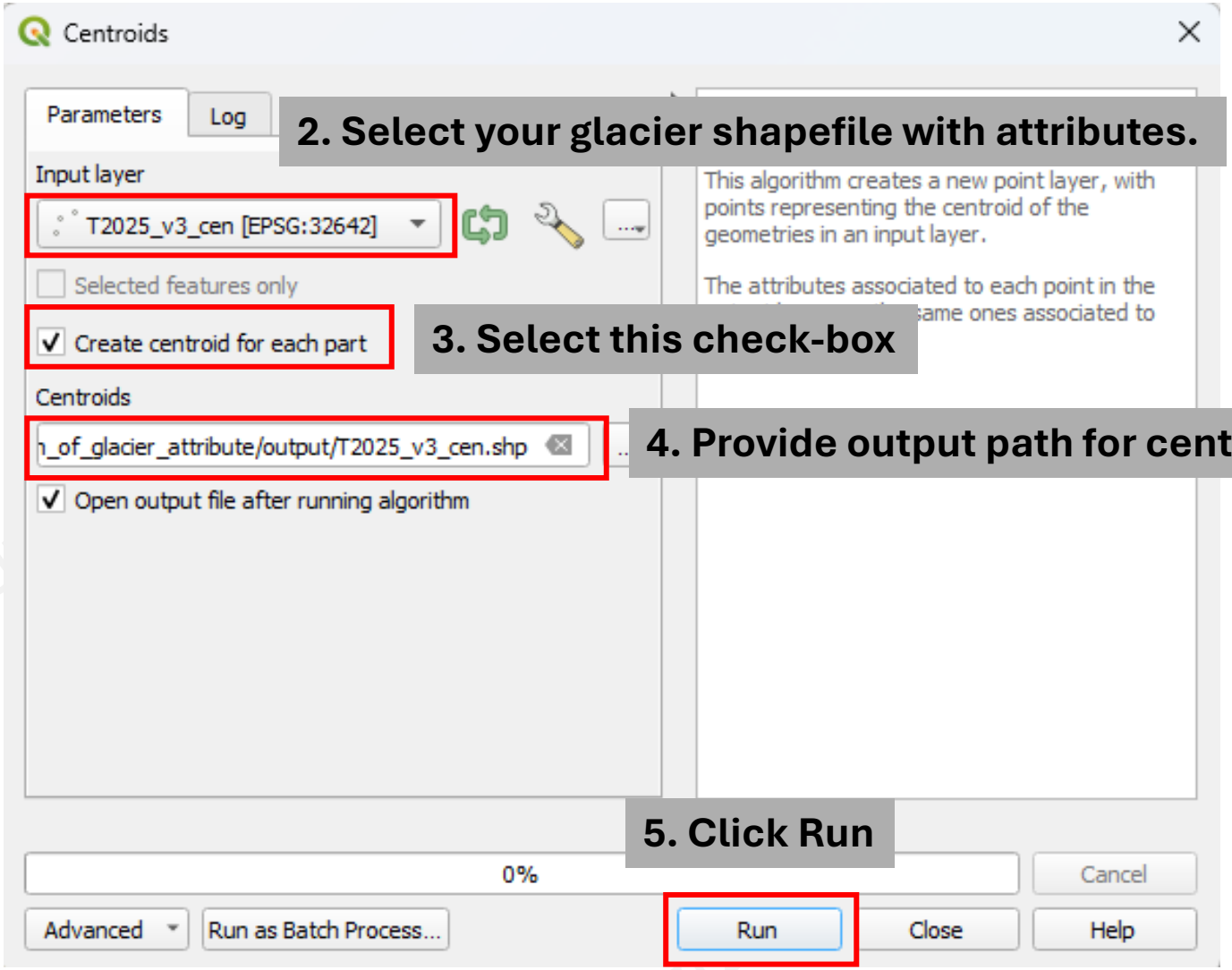
4. Apply ‘ Create a new field’
Output field name = ‘YEAR’
Output field type = ‘Integer’
Output field length = 10

5. 2025

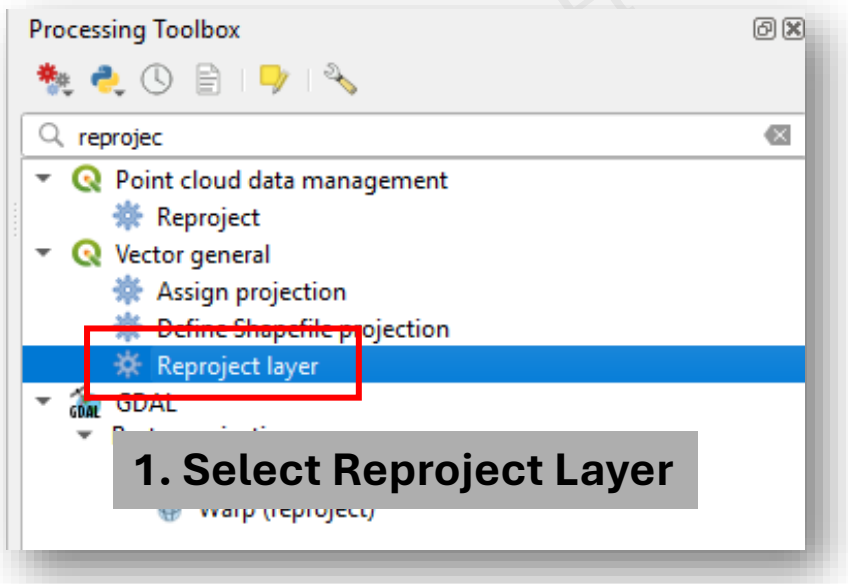
6. Click Ok and save the edit.

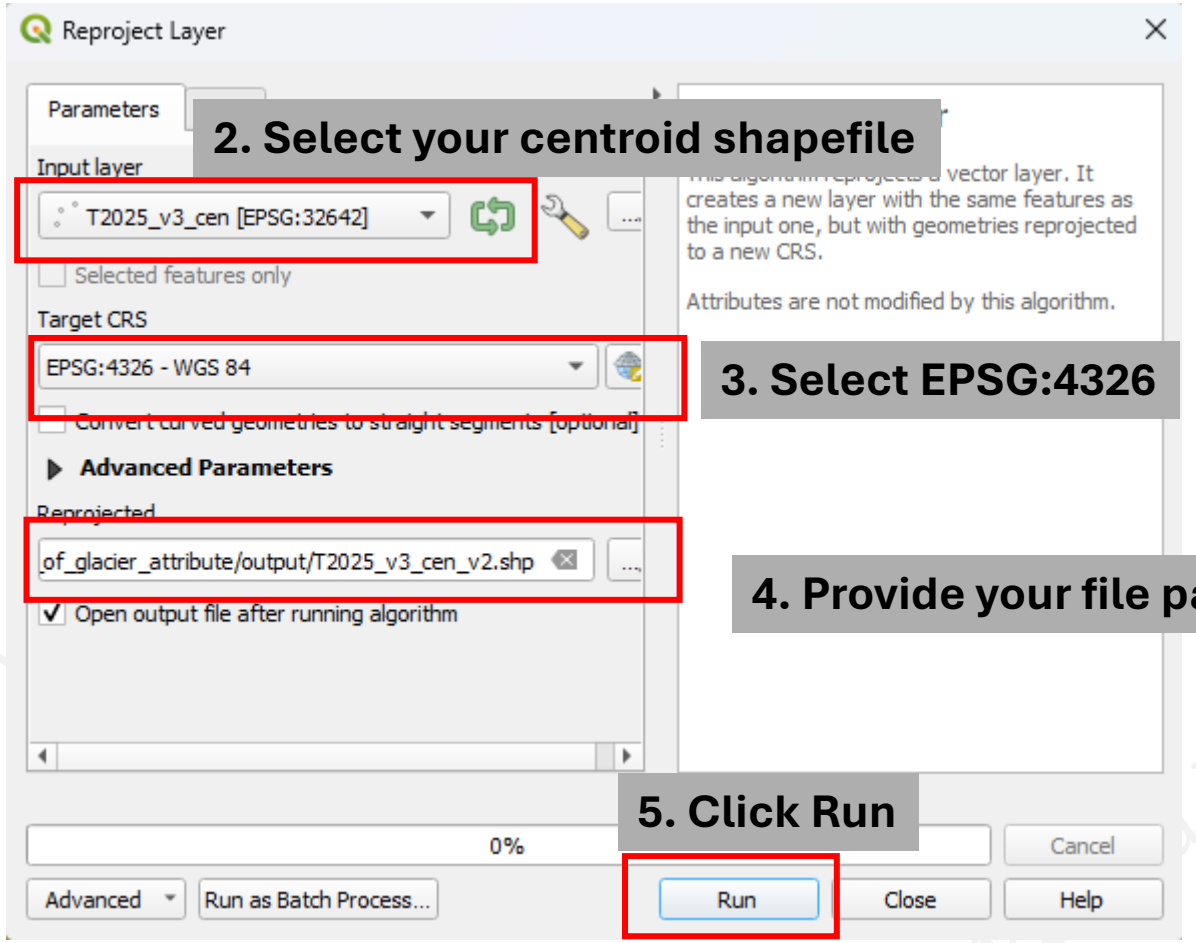
Centroids


Processing Toolbox
centroid
Centroids
Vector creation
Generate points (pixel centroids) along line
1. Select Centroids
Centroids
Point on surface
Pole of inaccessibility
GRASS
Vector (v.*)
v.delaunay
v.what.rast

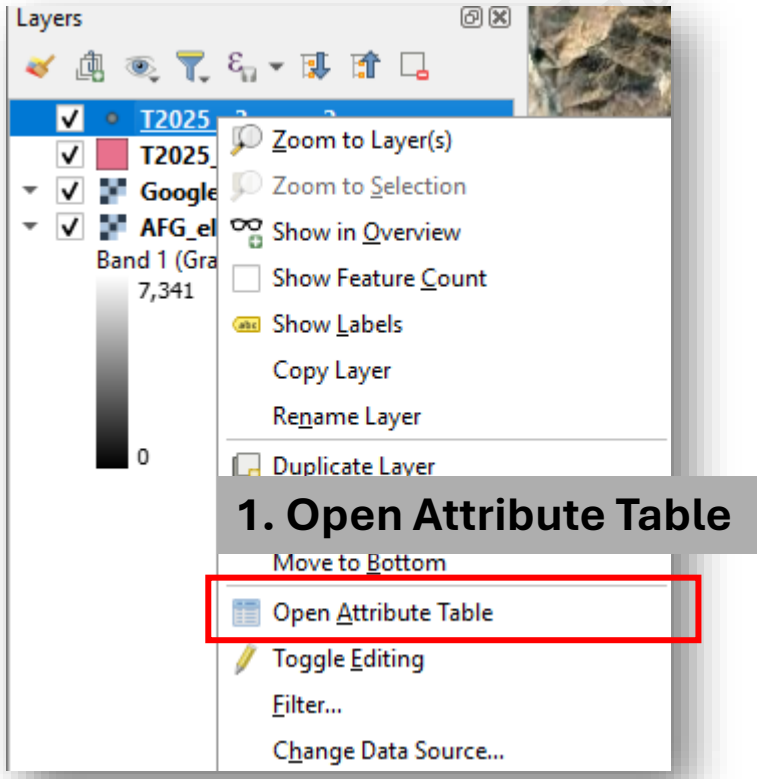

Centroids
Parameters
Log
2. Select your glacier shapefile with attributes.
Input layer
T2025_v3_cen [EPSG:32642]
This algorithm creates a new point layer, with points representing the centroid of the geometries in an input layer.
Selected features only
Create centroid for each part
3. Select this check-box
Centroids
_of_glacier_attribute/output/T2025_v3_cen.shp
4. Provide output path for centroids
Open output file after running algorithm
5. Click Run
0%
Cancel
Advanced
Run as Batch Process...
Run
Close
Help

Reprojection of Centroids into EPSG:4326 (GCS)

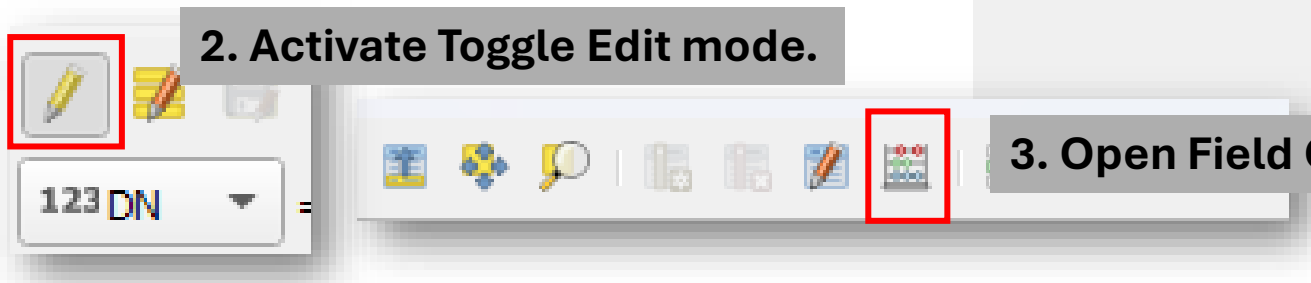




Centroids (Continued) Latitude

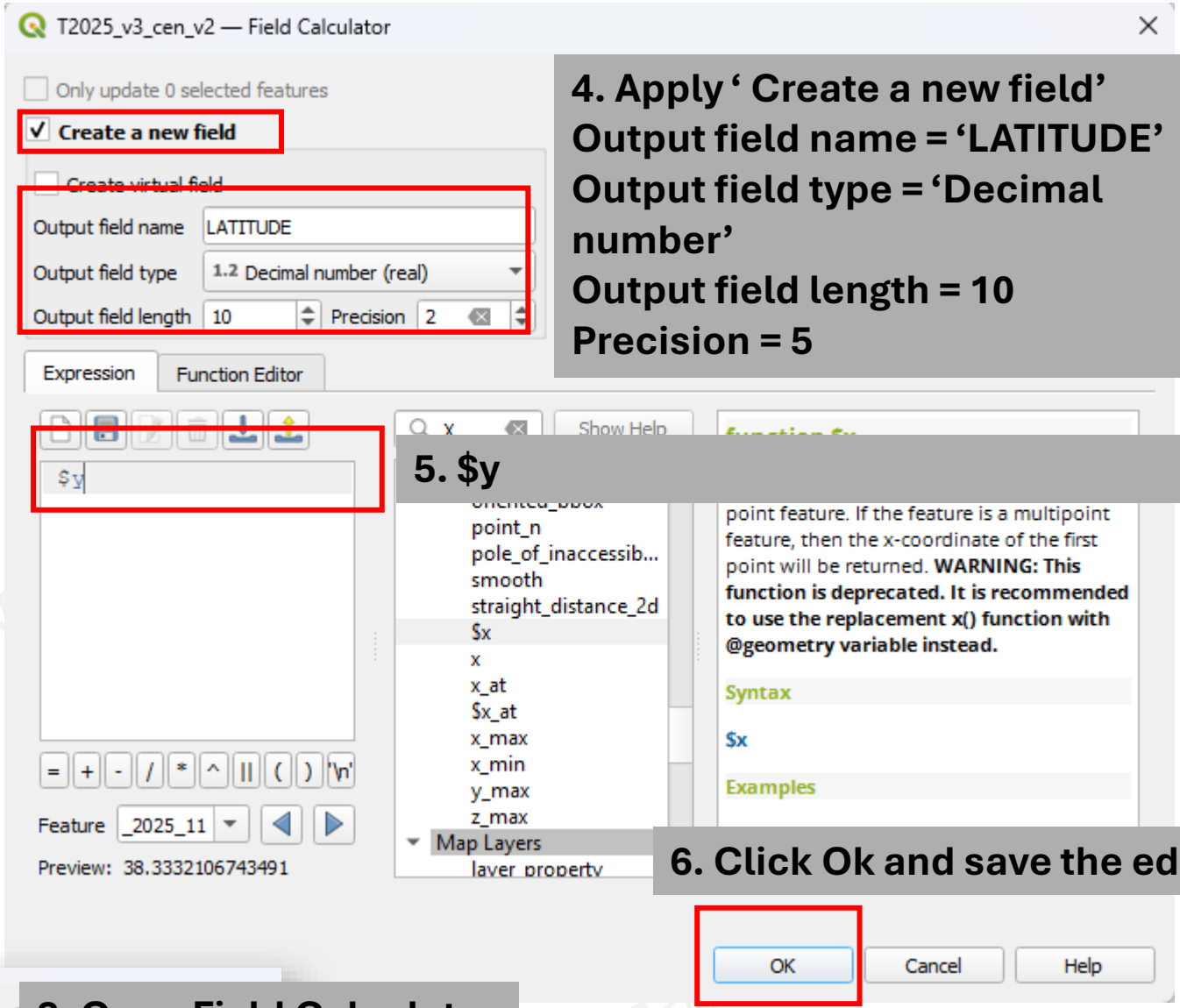


1. Open Attribute Table



2. Activate Toggle Edit mode.

3. Open Field Calculator

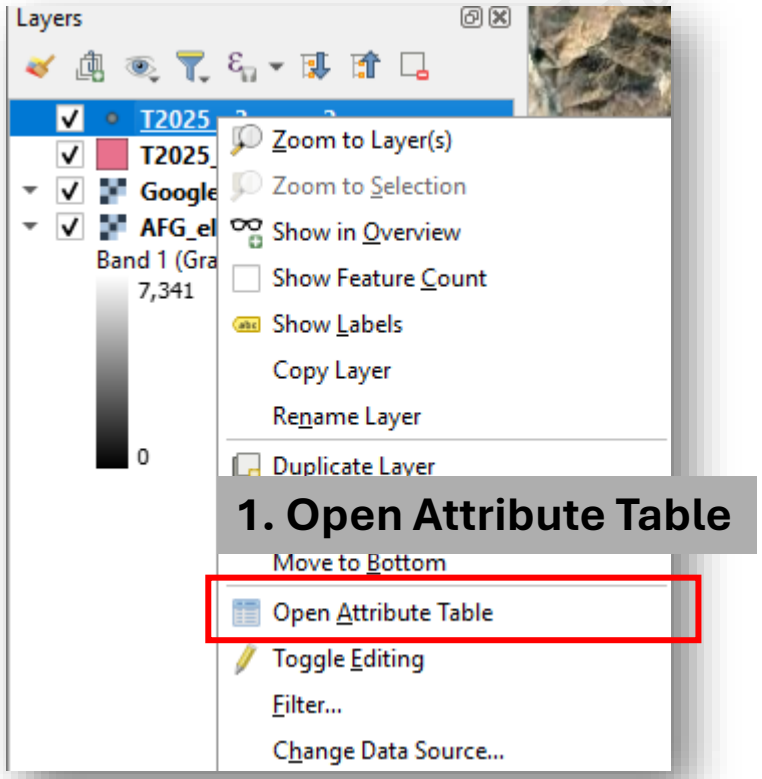


4. Apply ‘ Create a new field’
Output field name = ‘LATITUDE’
Output field type = ‘Decimal number’
Output field length = 10
Precision = 5

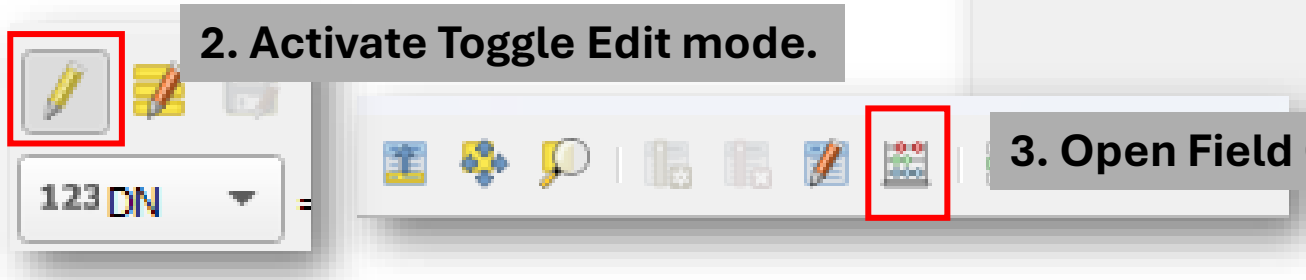
5. $y

6. Click Ok and save the edit.

Centroids (Continued) Longitude

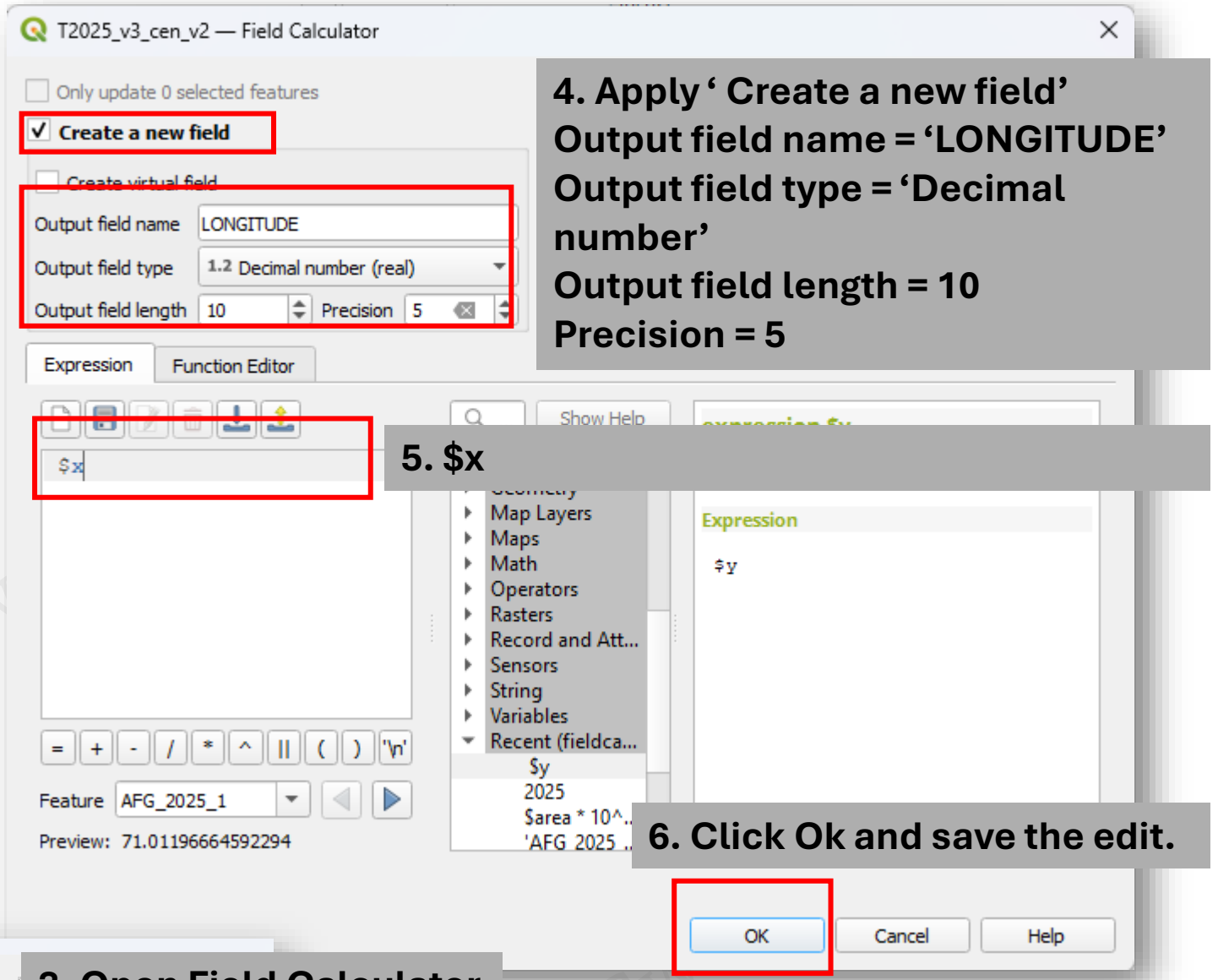


1. Open Attribute Table



2. Activate Toggle Edit mode.

3. Open Field Calculator

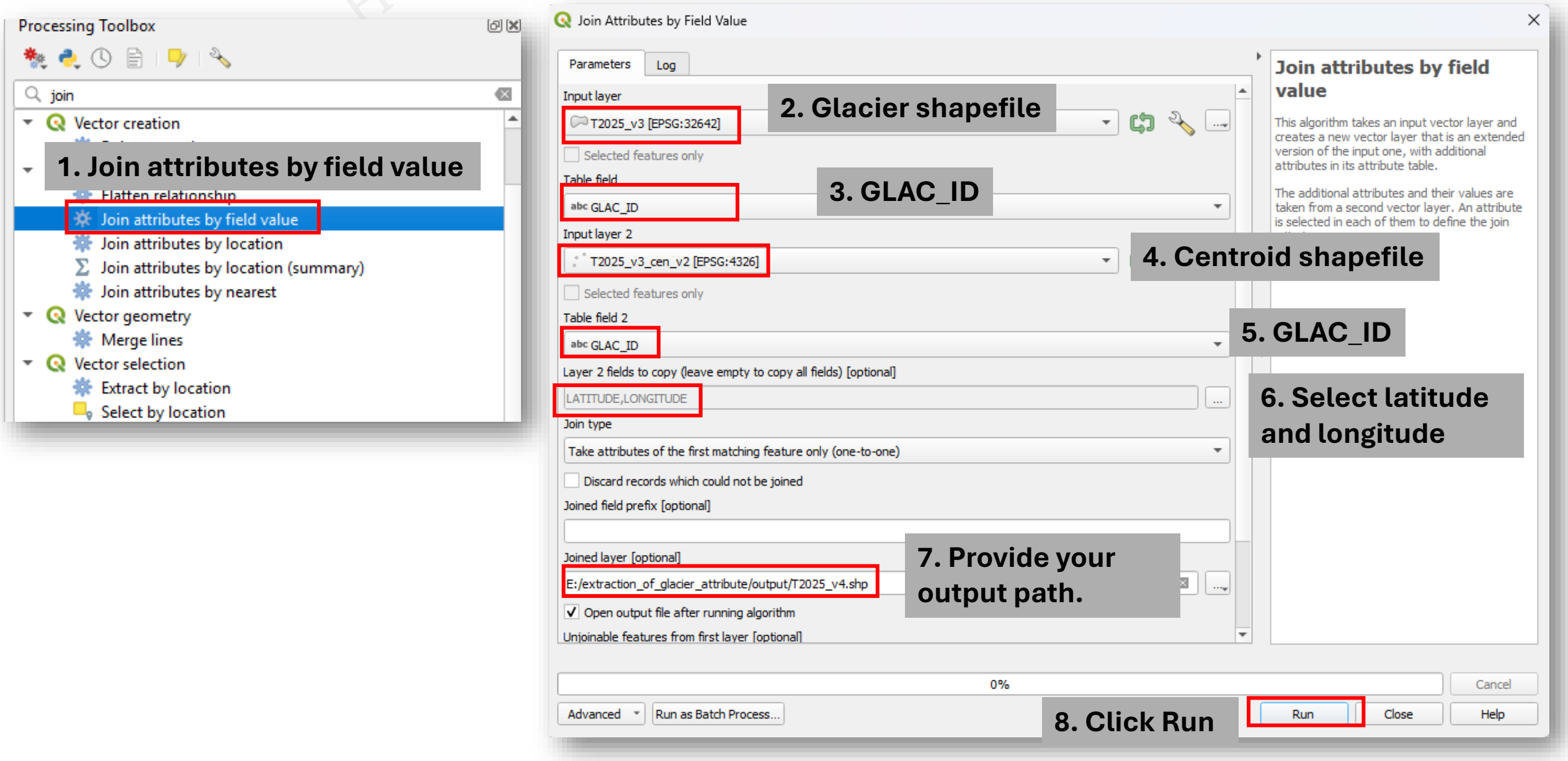


4. Apply ‘ Create a new field’
Output field name = ‘LONGITUDE’
Output field type = ‘Decimal number’
Output field length = 10
Precision = 5

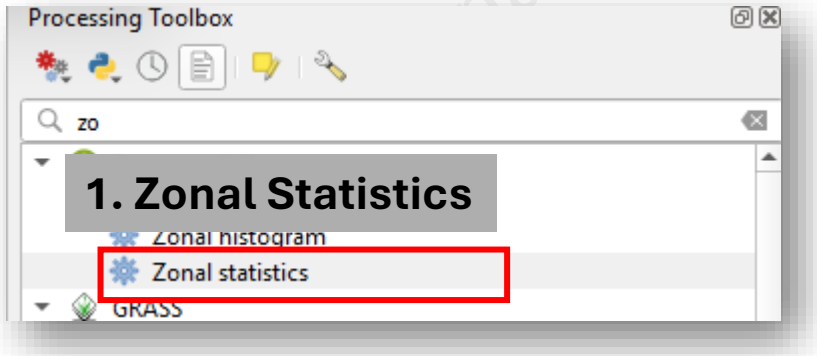
5. $x

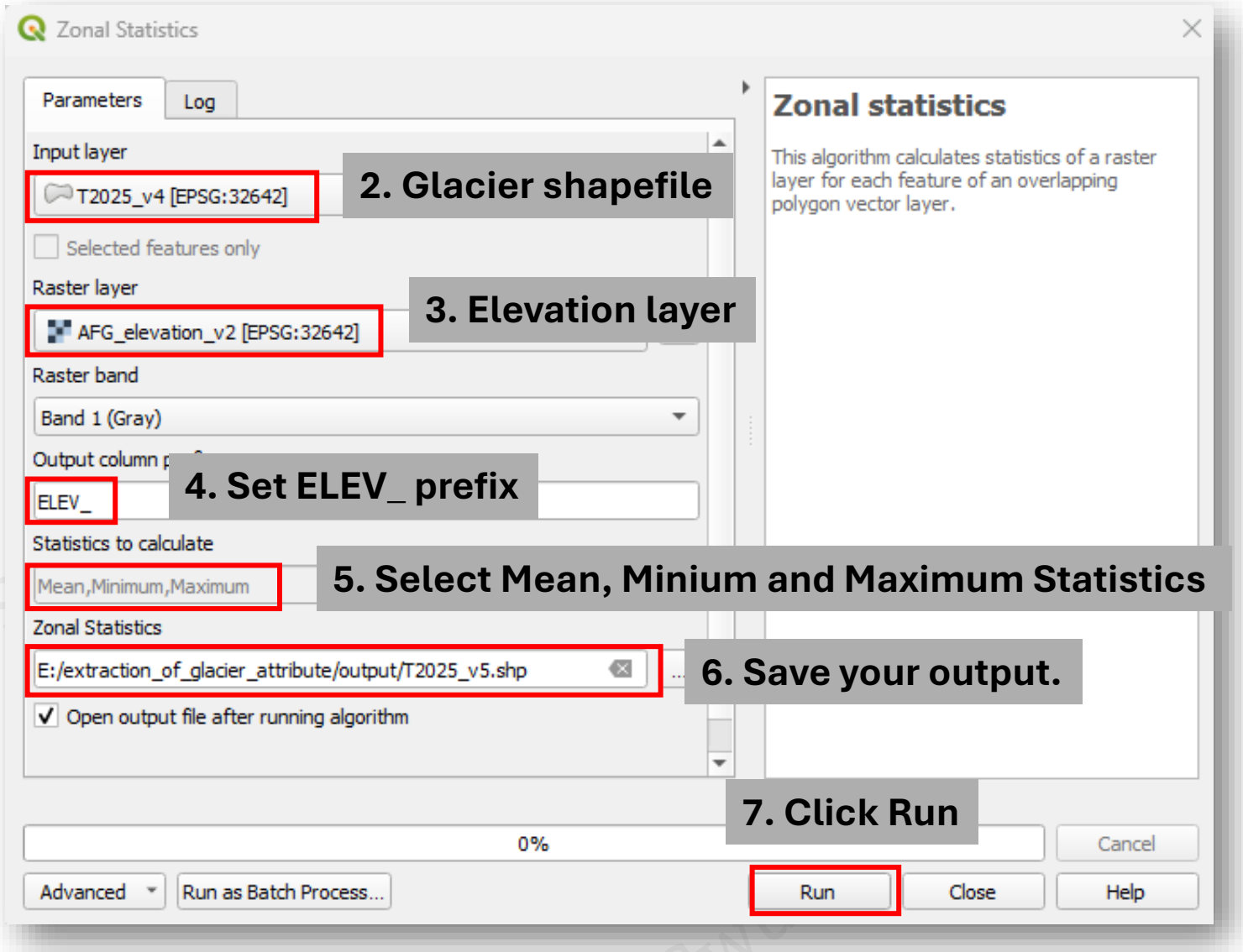
6. Click Ok and save the edit.

Joining centroid information with glacier polygons



Extraction of Mean, Minimum and Maximum Elevation Zonal Value





Note

- The same thing can be achieved for degree slope and aspect layers.
- The changed glacier polygons from glacier change analysis can also be used to extract layer zonal values.

Glacier Change Analysis

- Requires at least two years of acquisition: previous (X) and next (Y).

- Three types of changes:

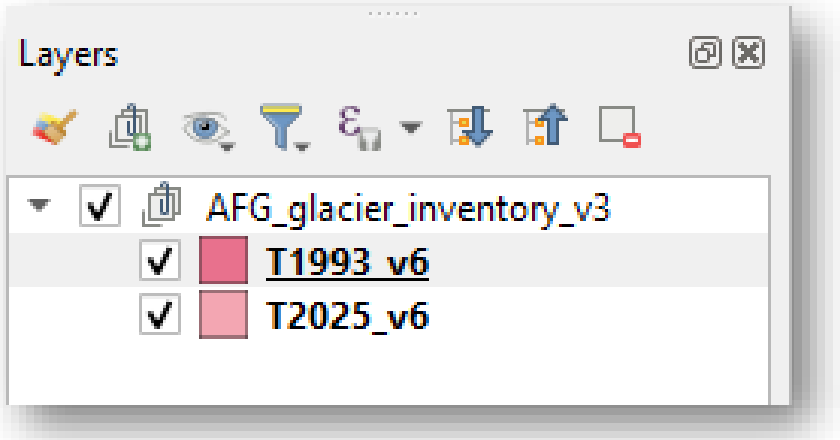
  - **Stable** = The glacier area retained throughout X to Y

  - **Advancing** = new glacier formation in Y (not seen in X)

  - **Retreating** = loss of glacier in Y (seen in X)

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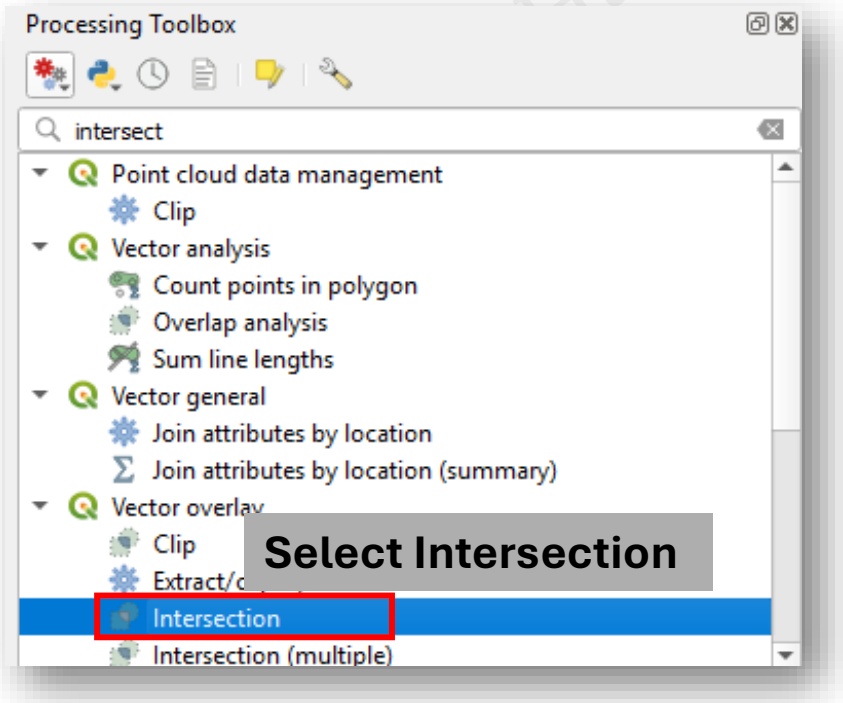
Vector Operations for Glacier Change Analysis

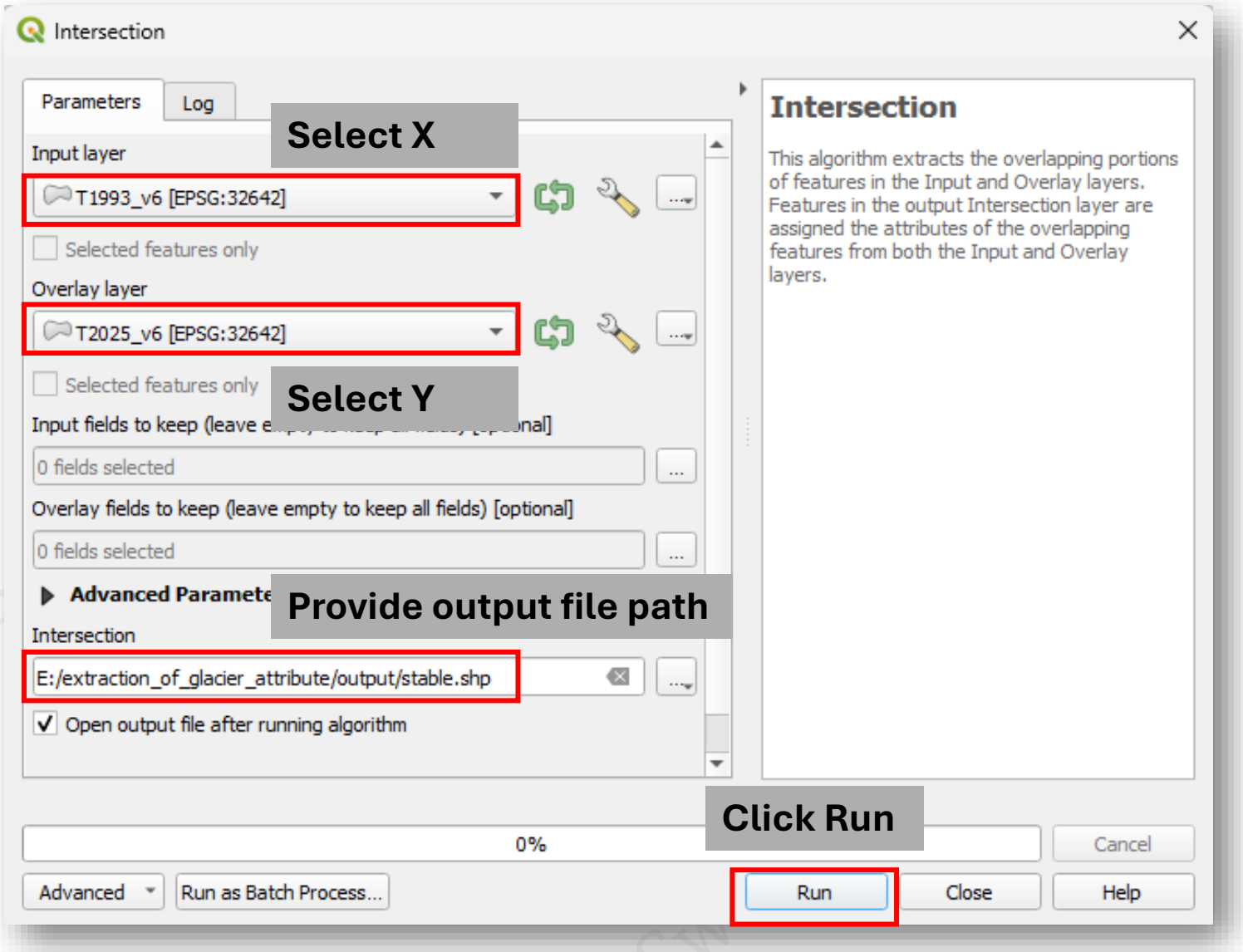


- Import glacier extents of X and Y to QGIS first.
- Three types of changes:
  - **Stable** = Intersection (X ∩ Y)
  - **Advancing** = Difference (X\Y)
  - **Retreating** = Difference (Y\X)

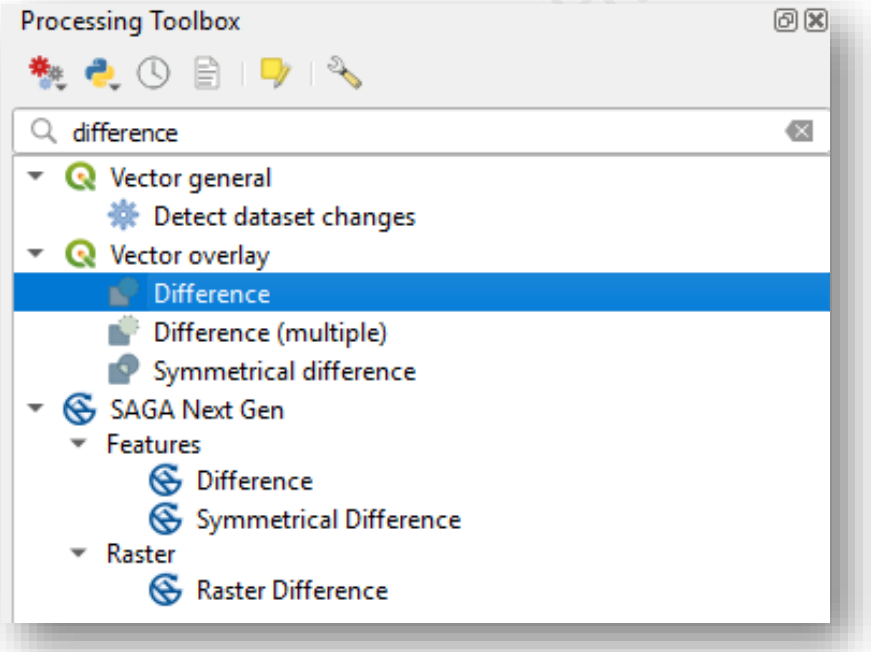
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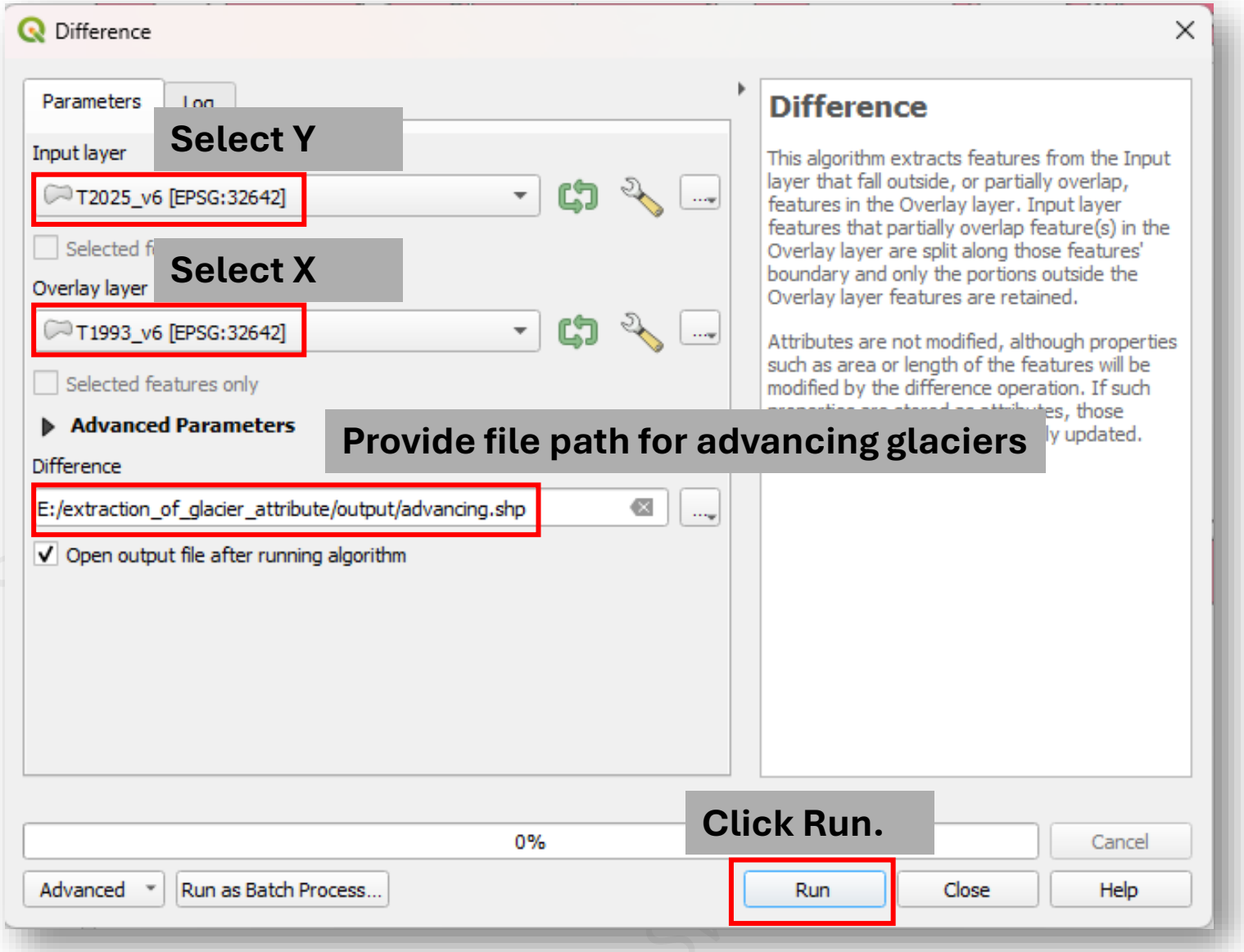
Stable Glacier Extraction



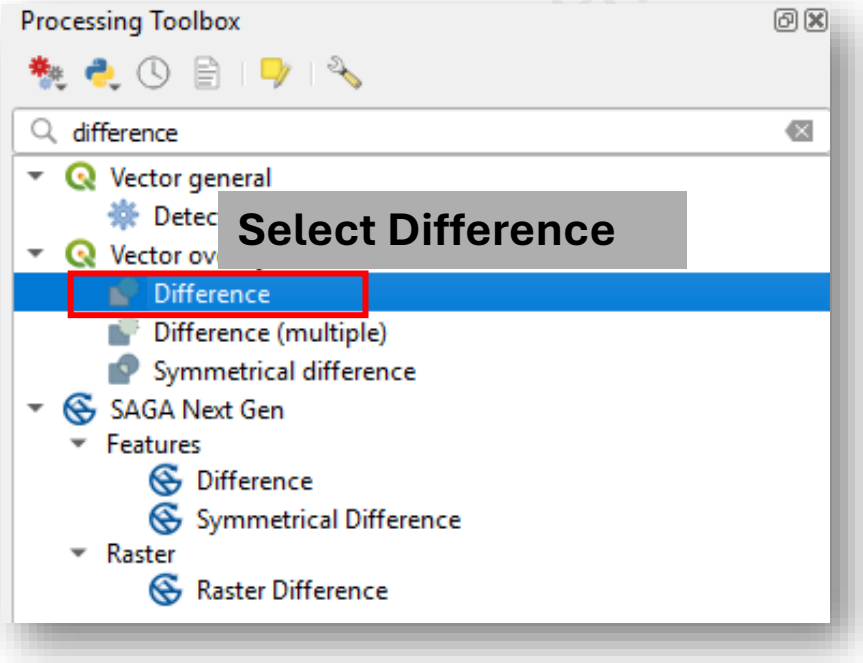


Advancing Glaciers





Retreating Glaciers



- From now, you can proceed further analysis with glacier change boundary data.

